

Number System Question Paper

02 July 2026 22:35



Maximum Marks: 50

Time: 1 Hour 30 Minutes

Instructions

1. Answer all questions.
2. Show all the conversion steps and calculations.
3. Use proper grouping methods for Binary-Octal and Binary-Hexadecimal conversions.
4. Scientific calculators are not allowed.

Section A - Number System Conversions (20 Marks)

Question 1 (Binary to Decimal Conversion) (4 Marks)

Convert the following binary numbers into their decimal equivalents.

- a) $(101101)_2$ (1)
- b) $(1110101)_2$ (1)
- c) $(110101.101)_2$ (1)
- d) $(10011.011)_2$ (1)

Question 2 (Decimal to Binary Conversion) (4 Marks)

Convert the following decimal numbers into binary.

- a) $(57)_{10}$ (1)
- b) $(125)_{10}$ (1)
- c) $(45.625)_{10}$ (1)
- d) $(23.375)_{10}$ (1)

Question 3 (Mixed Conversions) (6 Marks)

Convert the following:

- a) $(110111101)_2$ to Octal (1)
- b) $(572)_8$ to Binary (1)
- c) $(101101011)_2$ to Hexadecimal (1)
- d) $(9AF)_{16}$ to Binary (1)
- e) $(156)_{10}$ to Hexadecimal (1)
- f) $(3E9)_{16}$ to Decimal (1)

Question 4 (Advanced Conversions) (6 Marks)

Convert the following:

- a) $(4D7)_{16}$ to Octal (2)
- b) $(745)_8$ to Hexadecimal (2)
- c) $(A3F)_{16}$ to Octal (2)

Section B - Binary Arithmetic (14 Marks)

Question 5 (Binary Addition) (4 Marks)

Perform the following additions.

- a)
$$\begin{array}{r} 101101 \\ +110111 \\ \hline \end{array}$$
 (2)

b)

$$\begin{array}{r} 1110101 \\ +1011110 \\ \hline \end{array} \quad (2)$$

Question 6 (Binary Subtraction) (4 Marks)

a) Perform the subtraction using One's Complement Method.

$$101101_2 - 011011_2 \quad (2)$$

b) Perform the subtraction using Two's Complement Method.

$$1111000_2 - 0110101_2 \quad (2)$$

Question 7 (Binary Multiplication and Division) (6 Marks)

a) Multiply

$$1011_2 \times 110_2 \quad (3)$$

b) Divide

$$111100_2 \div 101_2 \quad (3)$$

Section C - Hexadecimal and Octal Arithmetic (16 Marks)

Question 8 (Hexadecimal Arithmetic) (8 Marks)

Perform the following operations.

a)

$$A7_{16} + 3D_{16} \quad (2)$$

b)

$$C4_{16} - 5B_{16} \quad (2)$$

c)

$$2A_{16} \times 4_{16} \quad (2)$$

d)

$$9C_{16} \div 3_{16} \quad (2)$$

Question 9 (Octal Arithmetic) (8 Marks)

Perform the following operations.

a)

$$657_8 + 245_8 \quad (2)$$

b)

$$732_8 - 165_8 \quad (2)$$

c)

$$37_8 \times 5_8 \quad (2)$$

d)

$$754_8 \div 6_8 \quad (2)$$

Mark Distribution

Section Topic		Marks
A	Number System Conversions	20
B	Binary Arithmetic	14
C	Hexadecimal & Octal Arithmetic	16
Total		50